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Me and My AI Bot: Exploring the 'AIholic' Phenomenon and University Students' Dependency on Generative AI Chatbots - Is This the New Academic Addiction?

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Abstract

Amidst the buzz of technological advancement in education, our study unveils a more disconcerting narrative surrounding student chatbot interactions. Our investigation has found that students, primarily driven by intrinsic motivations like competence and relatedness, increasingly lean on chatbots. This dependence is not just a preference but borders on an alarming reliance, magnified exponentially by their individual risk perceptions. While celebrating AI's rapid integration in education is tempting, our results raise urgent red flags. Many hypotheses were supported, pointing toward a potential over-dependence on chatbots. Nevertheless, the unpredictable outcomes were most revealing, exposing the unpredictable terrain of AI's role in education. It is no longer a matter of if but how deep the rabbit hole of dependency goes. As we stand on the cusp of an educational revolution, caution is urgently needed. Before we wholly embrace chatbots as primary educators, it is imperative to understand the repercussions of replacing human touch with AI interactions. This study serves as a stark wake-up call, urging stakeholders to reconsider the unchecked integration of chatbots in learning environments. The future of education may very well be digital, but at what cost to human connection and autonomy?

Introduction

In the annals of academic progression, the advent of artificial intelligence (AI) and its resultant tools have sparked both enthusiasm and introspection. As we steadily transition into a digital age marked by the omnipresence of AI, the educational landscape is undergoing an unprecedented transformation. Chatbots, emblematic of this AI revolution, have permeated classrooms, online courses, and student interfaces, promising a future of personalized learning experiences. Nevertheless, as with any technological boon, integrating chatbots into the educational matrix brings challenges and concerns (Salah, Alhalbusi, et al., 2023).

Terming the "AIholic" phenomenon, a burgeoning discourse surrounds students' psychological and behavioral patterns in their interactions with chatbots. While these AI-driven conversational agents initially emerged as ancillary tools, they now stand at the forefront of many educational settings, potentially reshaping student behavior, needs, and expectations. The allure of chatbots, fortified by their ability to simulate human dialogue, offer round-the-clock support, and cater to individualized learning trajectories, has raised poignant questions about dependency, compulsive behavior, and the intrinsic psychological needs they may fulfill or exploit (Kissinger et al., 2021; Sadiku et al., 2022).

This research ventures into the complexities of the "AIholic" dilemma. By employing rigorous methodologies and referencing foundational behavioral models, we scrutinize the multifaceted aspects of chatbot utilization in educational realms. We probe questions like: How do digital triggers shape chatbot dependency? How do intrinsic psychological drives, such as competence, relatedness, and autonomy, amplify students' reliance on these agents? Moreover, amid these intricate relations, where does risk perception fit in, influencing how students can regulate their chatbot interactions?

As we embark on this exploratory journey, we intend to provide a nuanced understanding of the AIholic phenomenon, bridging the technological strides with their psychological underpinnings. In doing so, we hope to offer insights that can guide educators, policymakers, and technologists in fostering a balanced and beneficial integration of chatbots in academic settings.

Literature Review

1. The Evolution of Chatbots in the Educational Landscape

The education landscape, traditionally rooted in brick-and-mortar classrooms and face-to-face interactions, has witnessed radical transformations with the advent of technology. The most prominent among these transformations is the integration of Artificial Intelligence (AI) into pedagogical methods, signaling a significant paradigm shift in how education is delivered and consumed (Alam, 2021; Roscoe et al., 2022; Salah, Abdelfattah, et al., 2023)

Central to this AI-driven revolution are chatbots. These conversational agents, designed to emulate human conversation using intricate algorithms, have swiftly transitioned from their nascent roles in domains like customer service to becoming integral

components of the educational framework (Dwivedi et al., 2021; Salah, Al Halbusi, et al., 2023). As these systems have grown more sophisticated, their potential applications in educational contexts have multiplied, capturing the attention of educators, administrators, and researchers alike.

In their early iterations within academia, chatbots were primarily perceived as supplementary tools, offering rudimentary support like answering frequently asked questions or guiding students through course registrations (Opoku-Brobbey, 2020). However, as their capabilities expanded, so did their roles. Institutions soon recognized the invaluable potential of chatbots to offer 24/7 support, drastically reducing the latency in addressing student queries and concerns, which traditional human-operated systems could not achieve (Ramirez, 2021).

Beyond just support, the promise of personalized learning tailored to individual student needs propelled chatbots to the forefront of educational innovations. Advanced chatbots, equipped with machine learning algorithms, began to analyze student interactions, adapt content delivery in real-time, and offer custom feedback, facilitating an unprecedented level of individual attention (Wang et al., 2023).

Furthermore, the global shift towards online learning, accelerated by external factors like the COVID-19 pandemic, underscored the indispensability of chatbots. In virtual learning environments, where traditional student-teacher interactions are sometimes constrained, chatbots emerged as consistent, reliable, and ever-present companions to students, guiding them through their academic journeys and ensuring they remain engaged and motivated (Chuang et al., 2023).

The meteoric rise of chatbots in academia is not merely a by-product of technological advancement. It is a testament to the evolving needs of modern learners and the relentless pursuit of the educational sector to adapt, innovate, and provide optimal learning experiences in a rapidly changing world.

2. Dual Facets of Chatbot Engagement: Promises and Pitfalls

The integration of chatbots into educational platforms paints a picture of progress contrasted by concerns. As the educational landscape has grown more complex, so has the role of these AI-powered tools, evoking a range of reactions from educators, researchers, students, and stakeholders.

Promises:

- **Individualized Learning Experiences**: The adaptability and scalability of chatbots set the stage for genuinely personalized educational experiences. Chatbots, with their underlying AI mechanisms, unlike traditional educational tools, can understand and cater to individual student preferences and learning paces. They can adjust the complexity of content based on student performance, ensuring each learner gets a tailored experience (Khalil & Rambech, 2022).
- **Immediate Feedback**: A significant advantage of chatbots lies in their ability to provide instantaneous feedback. Students no longer have to wait for periodic assessments or teacher evaluations. Instead, they receive immediate responses, helping them understand their areas of strength and aspects needing improvement, fostering a culture of continuous learning (Lechler et al., 2019).
- **Socio-Emotional Support**: As chatbot technology advanced, its scope expanded beyond academic support. Modern chatbots are designed to recognize emotional cues in user inputs, enabling them to offer socio-emotional support. In contexts where human support might be limited or delayed, chatbots serve as immediate, albeit virtual, support pillars, assisting students in managing stress and other emotional challenges (Ifelebuegu et al., 2023).

Pitfalls:

- **Data Privacy Concerns**: As with most digital tools, chatbots accumulate vast amounts of data, often personal and sensitive. This accumulation raises pressing concerns about data privacy and security. While most educational institutions ensure stringent data protection measures, the potential for breaches remains a looming concern, especially with third-party chatbot providers (Salah, Al Halbusi, et al., 2023).

- **Erosion of Human Touch**: The essence of education lies in content delivery and fostering human connections—between teachers and students and among peers. Over-reliance on chatbots might inadvertently dilute this essence, leading to a more mechanized and less empathetic learning environment. There is a growing debate on finding the right balance, ensuring that chatbots complement human interactions rather than replace them (Ifelebuegu et al., 2023).
- **Over-dependency on Technology**: An extension of the above point, there is a genuine concern that excessive use of chatbots might cultivate a culture of over-dependency on technology. Students might become too reliant on chatbot assistance, hampering their ability to seek solutions independently or collaborate with peers (Woithe & Filipec, 2023).

In sum, while chatbots undeniably offer numerous advantages, they also bring forth challenges that the educational community needs to navigate judiciously. Striking the right balance between leveraging the benefits and mitigating the pitfalls will define the future trajectory of chatbot integration in education.

3. Compulsive Behavior Theory and the "AIholic" Paradigm

While bringing unparalleled conveniences, the digital age has also ushered in unique challenges. One such challenge is the increasing dependency on technology, which, in extreme cases, mirrors patterns of compulsive behaviors. The Compulsive Behavior Theory (Jones & Menzies, 1997) provides a lens through which such patterns can be observed and understood, especially in emerging technologies like chatbots.

Understanding Compulsive Behavior Theory:

At its core, the Compulsive Behavior Theory postulates that certain individuals exhibit repetitive behaviors, often driven by specific triggers, even when such actions lead to adverse consequences. This pattern is not a mere habit but rather a compulsive need that the individual finds challenging to control or reduce (Luigjes et al., 2019).

Traditionally, this theory has been applied to gambling, eating disorders, and substance abuse. However, with the rise of digital technologies, researchers have started exploring its applicability to technology-related behaviors, including chatbots.

Chatbots and the Emergence of the "AIholic":

Due to their real-time response mechanisms and tailored interactions, Chatbots have become indispensable tools for many, especially students seeking instant information or feedback. However, the ease of access and the instant gratification they provide can also serve as triggers, as delineated by the Compulsive Behavior Theory.

The use of technology, researchers have identified potential triggers for compulsive chatbot use, ranging from emotional states, like loneliness or anxiety, to environmental cues, such as academic pressures or the sheer ubiquity of digital devices. For a student feeling overwhelmed, the immediate assistance of a chatbot can be enticing, leading to repeated and excessive use (Brubaker, 2022; Nosrati et al., 2020).

Hence, this paper argues that an "AIholic" individual exhibits an over-reliance on chatbot interactions, often at the expense of human interactions or other essential activities. This over-dependency can manifest in various ways: a student perpetually seeking validation from a chatbot, continuous interaction even when not required, or the inability to disengage from chatbot interfaces despite negative repercussions, such as reduced human interaction or compromised sleep.

Implications and The Path Forward:

The "AIholic" paradigm is a testament to the intricate relationship between humans and modern technology. While chatbots promise efficiency and personalization, the potential for over-dependency is real (Woithe & Filipec, 2023). As researchers and educators, the challenge lies in recognizing these patterns early and devising strategies that promote balanced and healthy interactions with chatbots while safeguarding students' well-being.

4. Self-Determination Theory (SDT) and Digital Dependencies

The digital revolution has brought forth a myriad of tools designed to enhance human experiences, and in this transformative landscape, Self-Determination Theory (SDT) offers a framework to understand the nuances of human motivation to technology. Conceived by (Ryan & Deci, 2000), SDT delves deep into the psychological needs of individuals and how the fulfillment or lack thereof influences behavior and well-being.

Core Tenets of SDT (Wang et al., 2019):

SDT identifies three central, innate psychological needs:

1. **Competence**: This need speaks to one's desire to experience mastery and effectiveness in their endeavors. When met, it fosters a sense of confidence and growth.
2. **Relatedness** reflects the universal desire to connect with others, belong, and feel understood by those around us.
3. **Autonomy**: Autonomy is about agency – the feeling that one's actions and decisions are self-endorsed and aligned with one's values.

Chatbots and the Fulfillment of Psychological Needs:

Digital platforms, especially chatbots, interface uniquely with these needs:

- **Competence**: Chatbots, designed to provide instant, accurate, and tailored responses, can significantly enhance a user's feeling of competence. For students, this translates to successfully finding answers, understanding complex topics, or navigating academic challenges with the aid of chatbots, thereby reinforcing their sense of achievement and efficacy.
- **Relatedness**: The modern, often isolated digital landscape sometimes creates voids in human-to-human interactions. With their seemingly "human-like" conversations, Chatbots can artificially fill these voids. They provide a semblance of companionship, understanding, and interaction, which can especially resonate with users who may feel isolated or marginalized.
- **Autonomy**: While chatbots might not directly cater to the need for autonomy, their role is more of an enabler. Chatbots empower users to make informed decisions by providing swift and efficient support, promoting a sense of autonomy in their academic or personal pursuits.

The Double-Edged Sword of Digital Dependencies:

While chatbots can satisfy these psychological needs, the mechanisms that make them effective also carry the risk of over-reliance or dependency. Hence, this convenience might transition from a beneficial tool to a crutch, subtly steering students into a heightened reliance on AI for their psychological well-being, thereby creating a potential digital dependency (Xie et al., 2023).

The intersection of SDT and digital tools like chatbots paints a fascinating picture of modern human behavior. As these tools become more ingrained in daily routines, understanding their impact on fundamental psychological needs will be paramount in guiding their ethical and practical use, ensuring that they augment, rather than replace, the rich tapestry of human experiences.

5. The "AIholic" Phenomenon: An Intersection of Technology and Psychology

The rise of AI in various sectors has often been greeted with awe and skepticism. A new term, "AIholic," may seem to capture the essence of this spectrum. While its coinage may seem playful at first, it encapsulates a pressing concern — the profound influence of AI on human behavior and dependency. The allure of AI, particularly in chatbots, offers convenience, personalization, and a semblance of human interaction. Nevertheless, beneath these benefits lies a subtle dance of technological seduction and human psychology.

Chatbots: Beyond Tools, Towards Companions:

Historically, tools were inert, serving a specific purpose. However, modern AI tools, especially chatbots, are designed to engage, adapt, and empathize. Chatbots' constant availability and unwavering "patience" make them ideal learning companions (Q. Jiang et al., 2022; Salah, Alhalbusi, et al., 2023). However, these attributes can also contribute to an overreliance or emotional attachment, pushing the boundaries of traditional tool-user dynamics (Laestadius et al., 2022)

Psychological Underpinnings of the AIholic Behavior:

The intertwining of psychological needs with chatbot use is pivotal in understanding AIholic behavior. For instance, a student seeking competence might find solace in a chatbot's consistent, error-free responses. Those yearning for relatedness might mistake the simulated conversation of a chatbot for genuine human interaction, especially in digitally isolated environments. These patterns, influenced by AI's capabilities and inherent human needs, culminate in the AIholic phenomenon, where technology is not just a tool but an integral part of one's psychological fabric (Ferreri et al., 2018; Ogilvie et al., 2022; Salah, Abdelfattah, et al., 2023)

As technology continues its relentless march forward, our understanding of its profound effects on the human psyche struggles to keep pace. Piecemeal research, which focuses on isolated aspects of chatbot usage or psychological impacts, has illuminated this intricate puzzle's specific facets. However, a holistic picture that seamlessly melds the technological marvels with their psychological implications remains elusive.

This research endeavors to bridge the scattered insights into a cohesive narrative. By juxtaposing the technological features of chatbots with theories from psychology, it seeks to unveil the deeper motivations, rewards, and possible pitfalls of AIholic behavior. In doing so, it aspires to guide educators, technologists, and policymakers in navigating the challenges and opportunities presented by this evolving phenomenon.

The Moderating Role of Risk Perception

The perception of risk is a multifaceted concept with roots in cognitive psychology, sociology, and behavioral economics. Risk perception addresses individuals' intuitive judgments and assessments regarding potential threats or dangers (Slovic, 1988). Recent literature has highlighted the significance of risk perception in influencing various behaviors, especially among university students (Leppin & Aro, 2009; Salah Hassan et al., 2021).

In the realm of addictive behaviors and dependencies, triggers, competence, relatedness, and autonomy are crucial variables. Recent studies on university students have shown that risk perception plays a pivotal role in moderating these relationships. Slovic (1988) explained that risk perception varies among individuals based on their experiences, knowledge, and cultural backgrounds. Consequently, how university students perceive risks associated with their behaviors, particularly concerning triggers and dependency, can differ widely (Sheeran et al., 2014).

In the realm of competence, Bandura's work on self-efficacy posits that when individuals perceive challenges as surpassing their coping abilities, their belief in their competence to handle such situations may diminish. This diminished self-efficacy, mainly observed among university students, can influence a decreased confidence in curtailing certain addictive behaviors or dependencies (Bandura, 1977; Hassan et al., 2022)

Relatedness and autonomy also demonstrate relationships with risk perception. In their self-determination theory, Deci and Ryan (2000) posited that feelings of relatedness and autonomy are fundamental for well-being. However, when students perceive higher risks, their feelings of relatedness might decrease, leading to isolation and, consequently, an inability to reduce use (Vallerand, 1997).

As for the relationship between autonomy and the actual dependency level, risk perception has a moderating role. Oei and Morawska (2004) found that when university students perceived higher risks but felt autonomous, their dependency levels were lower. Conversely, a perceived lack of autonomy combined with high-risk perception can lead to increased dependency levels.

In conclusion, risk perception is a significant moderator between triggers, competence, relatedness, autonomy, and outcomes related to dependency. Especially among university students, understanding the nuances of risk perception can offer insights into preventive measures and intervention strategies.

Research hypothesis:

1. Triggers and Dependency:

• Hypothesis 1

• There is a positive correlation between the frequency and intensity of triggers experienced by students and their actual dependency level on chatbots.

• Hypothesis 2

• The more frequently students experience triggers, the more challenging it becomes to reduce their use of chatbots.

2. Needs for Competence and Dependency:

• Hypothesis 3

• Students with a higher intrinsic need for competence show a greater dependency on chatbots.

• Hypothesis 4

• As students' need for competence increases, their ability to reduce chatbot use diminishes.

3. Needs for Relatedness and Dependency:

• Hypothesis 5

• Students with a more substantial need for relatedness show a more significant dependency on chatbots.

• Hypothesis 6

• The more substantial the need for relatedness in students, the harder it becomes for them to reduce their use of chatbots.

4. Needs for Autonomy and Dependency:

• Hypothesis 7

• A positive correlation exists between students' intrinsic need for autonomy and their actual dependency level on chatbots.

• Hypothesis 8

• As students' intrinsic need for autonomy rises, their ability to reduce chatbot use decreases.

The moderating effect:

Hypothesis 9

For university students, the correlation between triggers experienced and actual chatbot dependency level is influenced by their perception of risk. Notably, when risk perception is low, there is a more vital positive link between experienced triggers and dependency, but this relationship diminishes with higher risk perception.

Hypothesis 10

The association between triggers encountered by students and their struggle to limit chatbot usage is moderated by their sense of risk. In essence, the affirmative tie between triggers and reduced usage is intensified in students with a lower-risk

outlook while being milder in those perceiving higher risks.

Hypothesis 11

Risk perception shapes the link between students' need for competence and their reliance on chatbots. This positive connection between the quest for competence and dependency amplifies for those with a lesser risk perception but weakens with heightened risk awareness.

Hypothesis 12

For students, the relationship between their competence needs and their challenge in curtailing chatbot engagement is tempered by their risk perception. The bond between competence needs and reduced engagement is more pronounced for those with lower perceived risks and less for those sensing more significant risks.

Hypothesis 13

Their risk views modulate the connection between students' yearning for relatedness and actual chatbot dependency. This affinity between the desire for relatedness and dependency grows more vital for students with minimal risk awareness but attenuates with increased risk consciousness.

Hypothesis 14

Their risk perspective regulates the interaction between students' drive for relatedness and their hurdle in minimizing chatbot interaction. This bond between the quest for relatedness and reduced interaction deepens for those perceiving lower risks and softens for those with a more cautious approach.

Hypothesis 15

Risk perception navigates the relationship between students' autonomy needs and genuine dependence on chatbots. Students with a subdued risk perception show a pronounced positive linkage between autonomy needs and dependency, while this link lessens for those with an acute sense of risk.

Hypothesis 16

Their risk viewpoint identifies a student's aspiration for autonomy and difficulty tapering chatbot utilization. A more potent positive correlation between autonomy aspiration and usage reduction is evident for those with diminished risk views, whereas it is less potent for those with heightened risk consciousness.

Methodology

Research Design

To comprehend the complexities of the "AIholic" phenomenon concerning chatbot usage among students, our research is designed to be quantitative and analytical. We will employ Structural Equation Modeling (SEM) using SmartPLS to test our hypotheses and model relationships between variables derived from the psychological theories. The advantage of using SEM is its capacity to assess complex models that comprise multiple dependent and independent variables.

Method and Materials

Structural equation modeling (SEM) was applied using the SmartPLS program (Henseler et al., 2015) to test the hypotheses. SEM—a powerful tool for examining the relationships among variables—offers several advantages over traditional regression methods (Hair, Sarstedt, et al., 2017; Hair Jr et al., 2021). This study used an online survey to collect data from university students who use chatbots or AI or any generative AI tools such as ChatGPT and Bard.

Measures

Several validated instruments were adopted from previous studies to measure the variables of interest and modified to fit the research goals. All items were rated on a 5-point Likert scale, ranging from "strongly disagree" to "strongly agree."

One academic expert in related fields reviewed and validated the questionnaire before the primary data collection phase. All scales' English versions were translated to Arabic using back translation—that is, translating the scales from English to Arabic and then translating them back from Arabic to English—to ensure the translation's accuracy. A bilingual expert reviewed the translated scales afterward to ensure their equivalence with the original scales.

Primary Data Collection: Methodology and Approach

A primary data collection approach was adopted to grasp the research questions' depth and nuances. An online survey tailored to the specific dimensions of our research served as our primary instrument.

Survey Instrument and Adaptation: The survey was meticulously designed, incorporating questions adapted from well-established and validated scales. These scales were pivotal in measuring our target variables: triggers for chatbot usage, the intrinsic needs for competence, relatedness, and autonomy, and the perceived inability among students to reduce or control their chatbot interactions.

Sample and Distribution: The survey targeted students from three distinct academic institutions - Karbala University, Al Safwa University College, and Al Anbar University. These institutions were selected to ensure a diverse and comprehensive representation of students, enhancing our findings' robustness and generalizability.

The survey link was disseminated via popular messaging platforms - WhatsApp and Telegram- to achieve a broad reach and facilitate easy access for participants. With their extensive user base among the student community, these platforms ensured that the survey reached a vast audience relatively quickly.

Response and Timeline: The response to the survey was encouraging. Three hundred sixty-six students participated and provided their insights, contributing to a rich dataset for our analysis. Data collection commenced on 4 May 2023 and concluded on 27 September 2023, giving ample time for students to respond, ensuring that the data collected was reflective and comprehensive.

Instrumentation

1. Triggers (Adapted from the Internet Addiction Test - IAT) Derived from Young's Internet Addiction Test (IAT), the measure for "Triggers" captures the stimuli or conditions prompting the use of AI chatbots. While the original IAT assessed triggers leading to internet usage, this study has adapted specific items to encapsulate triggers related to chatbot interactions. Participants must express their agreement using a 5-point Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree).
2. Inability to Reduce Use (Adapted from the Compulsive Internet Use Scale - CIUS) This dimension leverages items from the Compulsive Internet Use Scale (CIUS) to discern the challenges participants face when decreasing their interaction with chatbots. The original CIUS examined compulsive behaviors related to the internet. However, for this research, the focus has been shifted to ascertain the compulsive engagement with chatbots. Responses are collected on a 5-point Likert scale, from 1 (Strongly Disagree) to 5 (Strongly Agree).
3. Actual Dependency Level (Adapted from the Smartphone Addiction Scale - SAS) To understand the degree of dependency on chatbots, this measure takes its foundation from the Smartphone Addiction Scale (SAS) by Kwon et al. (2013). Though initially constructed for smartphone addiction, the scale's items have been adapted to mirror chatbot dependency's behavioral and cognitive aspects. Participants will indicate their concurrence to statements using a 5-point Likert scale, from 1 (Strongly Disagree) to 5 (Strongly Agree).

4. Need for Competence, Need for Relatedness, and Autonomy: to measure these constructs in the context of chatbot interactions, we have employed the Basic Psychological Needs Scale (BPNS). The BPNS, designed initially to gauge these three needs across various life domains, was adapted to probe specifically into fulfilling these needs through chatbot usage.

Data Analysis

Internal Reliability and Convergent Validity.

We undertook an in-depth analysis covering individual item reliability, internal consistency reliability, convergent validity, and discriminant validity. Evaluating item reliability, most items met or surpassed the advised level of 0.707 based on Hair et al., 2017 or were beyond the 0.5 benchmark, indicating a moderate to strong correlation with their respective constructs (Hulland et al., 2018). This is consistent with the criteria set by Hair et al. (2010). Notably, two items - TRG11 and IBRU1 - had loadings under 0.7, but they were retained as their presence did not undermine the integrity of the measurements for the associated primary or secondary constructs. We employed composite reliability to assess our constructs' internal consistency, which fluctuated between 0.751 and 0.926, comfortably above the suggested 0.70 benchmark (Hair et al., 2017). Further validating the convergent aspect, our constructs' average variance extracted (AVE) spanned from 0.533 to 0.767, above the recommended 0.5 benchmark (Hair, Sarstedt, et al., 2017; Hassan et al., 2021). Refer to Table 1 for details.

Regarding discriminant validity, we found it to be well-established. The Average Variance Extracted (AVE) of every construct was more significant than the shared variance with other latent variables, corroborating Hair et al. (2017) recommendations (See Table 2 for details). Additionally, following the guidance of Henseler et al. (2015), we employed the heterotrait-monotrait ratio (HTMT) of correlations, a method derived from the multitrait-multimethod matrix. Table 3 displays that the HTMT values remained consistently under 0.90, signifying robust discriminant validity between variable pairs. Notably, all HTMT values distinctly differed from 1, and their 95% confidence intervals (CI) did not include 1. This is in harmony with the standards set by Henseler et al. (2015), further attesting to the discriminant validity of the variables.

Theoretical Model Hypothesis Testing

In the following discourse, we delve into the detailed outcomes of our formulated hypotheses, spanning from H1 to H8. Our analysis has afforded some enlightening revelations about the direct effects of these hypotheses on the associated constructs.

Starting with H1, our findings showcased a significant relationship between triggers and the inability to reduce usage. This was substantiated by an impressive effect size of $\beta = 0.386$, further reinforced by a t-value of 6.467 and a notable p-value of less than 0.000. This led us to accept H1 confidently.

On the trail of H1, our exploration of H2 pointed towards a discernible effect of triggers on the dependency level. With $\beta = 0.198$ and a t-value of 3.019, all indications indicated a significant result, especially with a p-value lying comfortably below 0.001, validating the support for H2.

As we proceeded to H3, the data reflected a distinct correlation between competence and the inability to reduce usage. Here, the results manifested in $\beta = 0.280$ and a t-value of 3.990, coupled with an exceedingly significant p-value, cementing the confirmation of H3. Mirroring this positive trend, competence's influence on actual dependency levels, as postulated in H4, was solidified with $\beta = 0.301$, a t-value of 4.881, and an unequivocally significant p-value.

Transitioning our focus to H5, the dimension of relatedness surfaced as a potent predictor of the inability to reduce use. This was evidenced by the robust effect size of $\beta = 0.154$, a high t-value of 3.412, and a p-value less than 0.000, culminating in the validation of H5. Conversely, hypothesis H6, which postulated a connection between relatedness and actual dependency level, did not fare as well. The data, characterized by $\beta = 0.101$ and a t-value of 1.024, alongside a p-value of 0.153, led to H6 being unsupported.

As we approached the end of our hypothesis spectrum, H7 drew our attention to the interplay between autonomy and the inability to reduce use. The hypothesis stood firm with an effect size of β = 0.128 and a t-value of 2.583, especially with a p-value echoing significance. Finally, our scrutiny of H8 brought to light autonomy's marked relationship with the actual dependency level, backed by β = 0.242, a t-value of 4.672, and a highly significant p-value, sealing the confirmation of H8.

To facilitate a comprehensive grasp of these findings, readers are directed to Table 4 for a detailed breakdown and to Fig. 2 for a visual elucidation.
In line with the primary objectives of this study, the moderation analysis played a pivotal role in determining whether risk perception independent components, i.e., triggers, competence, relatedness, and autonomy, and the outcomes variables, i.e., inability to reduce use and actual dependency level. As mentioned, we employed the PLS bootstrapping method with 5,000 re-samples to analyze the structural model and generate t-values. The results of the moderation analysis for the interactions. All these results are presented in Table 5 and

Generally, it is unclear how a moderation analysis differs for high and low interaction. In other words, the size of the precise nature of this effect is not easy to define from the analysis of the coefficient itself (Dawson, 2014). Thus, Dawson (2014) suggested that this can be followed up with an interaction plot.

Assessment of the Predictive R^2

The coefficient of prediction (R2 value) indicates the model's prediction performance measured as the squared correlation between a particular endogenous component's actual and anticipated values. Furthermore, this coefficient represents the total number of the exogenous constructs' impacts on the given endogenous construct. This coefficient's value goes from 0 to 1, with more significant numbers indicating higher degrees of predicted accuracy. R^2 determines the overall effect of the model. In other words, R^2 is used as an indicator of the overall predictive strength of the model, and the rule of thumb, according to Hair, Hult, et al. (2017), is to cut off R^2 as follows:

- R^2 0.75 → Substantial
- R^2 0.50 → Moderate
- R^2 0.25 → Weak

Regarding the power of explanation, the model explains values of R-square 0.532 for the inability to reduce the use and 0.530 for the actual dependency level, thus indicating a moderate-to-significant effect (Hair et al., 2017).

Results and Discussion

1. Triggers and Actual Dependency on Chatbots:

Upon analyzing the collected data, we uncovered a notable positive correlation between the triggers and dependency on chatbots. This indicates a scenario where students increasingly rely on these digital interlocutors when confronted with stimuli or circumstances that nudge them toward utilizing AI chatbots. This relationship suggests a mechanism where the frequency and intensity of triggers directly influence the depth of chatbot dependency (Elliott, 2019).

This dependency is not just a mere inclination to use chatbots more often; it potentially indicates a deeper psychological reliance (Sugumar & Chandra, 2021). Just as individuals might be predisposed to engage in certain behaviors in response to specific emotional or environmental cues, students, it appears, might be turning to chatbots as a coping or adaptive mechanism in response to their triggers (Adam et al., 2021).

Interestingly, this observation is not isolated but finds resonance with prior academic endeavors. For instance, certain emotional states, ranging from loneliness to stress or specific environmental cues, such as academic pressures or social isolation, could serve as potent catalysts, driving individuals toward compulsive behaviors (Knack et al., 2020; Sedikides et al., 2015). Drawing parallels, it seems chatbots, with their round-the-clock availability and predictable responses, might provide

students with a semblance of stability or control in the face of these triggers. The findings suggest that similar undercurrents might shape chatbot interactions, underscoring the intricate interplay between technology and human psychology (Lexcellent, 2019).

2. Need for Competence and Actual Dependency on Chatbots:

When students perceive a tool or platform as an avenue to hone their skills or validate their competence, they naturally gravitate towards it more. Our findings suggest that chatbots play a pivotal role in this dynamic. The positive correlation between the intrinsic need for competence and actual chatbot dependency highlights an intriguing pattern: students are not just using chatbots for transactional exchanges. However, they are deeply intertwined in a relationship where they derive a sense of self-worth and validation.

This aligns with the fundamental tenets of the Self-Determination Theory, which emphasizes the importance of feeling competent and effective in interactions with the environment. For students, chatbots appear to be fulfilling this intrinsic need by offering instantaneous feedback, answering queries, or even guiding them through complex problems. Such interactions potentially bolster their confidence, making them feel more adept and knowledgeable (Racero et al., 2020; Soenens & Vansteenkiste, 2011).

What is particularly fascinating is how technology, especially chatbots, has seamlessly integrated into this psychological fabric. It is not merely about getting a task done but about affirming one's capabilities (Chandra et al., 2022). Moreover, as chatbots provide consistent, non-judgmental feedback, they inadvertently nurture this dependency. As students continually seek out these affirmations, their reliance on chatbots grows, showcasing a delicate dance between technological advancement and human intrinsic needs (Brandtzaeg & Følstad, 2018; Ifelebuegu et al., 2023).

3. Need for Relatedness and Actual Dependency on Chatbots:

Amidst the ever-evolving digital landscape, connection and belonging remain paramount. Our analysis sheds light on an intriguing facet of human-AI interaction; students are not solely seeking informational or transactional exchanges with chatbots and a semblance of connection and interaction. The positive correlation between the intrinsic need for relatedness and chatbot dependency underscores this burgeoning dynamic (Xie et al., 2023).

Self-determination theory has always emphasized the innate human desire to form connections and feel a sense of belonging. The rise of digital platforms and AI tools like chatbots brings a unique twist to this. Traditionally, relatedness was sourced from human-to-human interactions, and the current trend suggests a drift towards human-AI relatedness. In scenarios where human interactions might be sparse, distant, or even impersonal, chatbots offer a consistent, always-available conduit for interaction, making them an appealing alternative (Scherer & Candrian, 2023).

Furthermore, chatbots' programmed responsiveness and lack of negative emotional responses might give some users a 'safe space.' Students might find solace in these interactions' predictability and non-judgmental nature, amplifying their reliance on them. This phenomenon prompts a broader reflection on the evolving nature of relatedness in the digital era, where algorithmic interactions could increasingly satiate our intrinsic social needs (Ta et al., 2020).

4. **Autonomy and Actual Dependency on Chatbots**: Our data analysis delineated a notable positive relationship between the intrinsic need for autonomy and actual dependency on chatbots. This suggests that students who value autonomy in their learning processes and perceive chatbots to exercise this independence are more prone to develop a dependency on these AI interfaces. Such findings can be grounded in the notion that chatbots empower students with immediate, personalized, and unmediated access to information, allowing them to govern their learning trajectories. This, in turn, intensifies their reliance on chatbots, reinforcing the significance of autonomy in shaping digital dependencies in the educational realm (Jiménez-Barreto et al., 2021; Xia et al., 2023).

5. Triggers and Inability to Reduce Chatbot Use:

The nexus between triggers and the escalating challenge to curtail chatbot usage offers a profound insight into the compulsive nature of digital interactions. In the context of our findings, it is evident that specific stimuli, whether they be emotional states, environmental cues, or other triggering factors, amplify the gravitational pull toward chatbots (McGinn, 2020; McStay, 2018).

Compulsive Behavior Theory provides a lens through which we can understand this dynamic. Just as specific triggers can lead individuals to substance addiction or other compulsive behaviors, the digital realm has its stimuli that can catalyze and reinforce addictive patterns. In this case, the digital 'substance' is the chatbot interaction (Huang & Bargh, 2014; Roth et al., 2021).

When students are repeatedly exposed to these triggers, the interaction cycle with chatbots becomes more entrenched. Over time, this can evolve into a habitual response, making disengaging from chatbots increasingly challenging. Chatbot responses' consistency, immediacy, and predictability might further entrench this behavior, creating a feedback loop that reinforces dependency (Moore & Hübscher, 2021).

This relationship between triggers and the inability to reduce chatbot use underscores the need for a more profound understanding of the stimuli leading to such digital compulsions. As educational institutions and policymakers seek to harness the potential of AI-driven tools, it becomes imperative to recognize and mitigate the risks associated with over-reliance and compulsive engagement.

6. Need for Competence and Inability to Reduce Chatbot Use:

The evident correlation between the intrinsic need for competence and the inability to reduce chatbot interactions offers profound insights into the complexities of human-chatbot dynamics within educational settings. This correlation suggests that students, driven by their inherent desire to master specific skills and knowledge, might perceive chatbots as a valuable tool that validates their competencies (Nguyen, 2021; Pentina et al., 2023).

However, this validation might come at a cost. While chatbots can effectively provide instant feedback, answer queries, and augment learning, over-reliance indicates a potential double-edged sword. On one hand, chatbots are meeting the students' intrinsic needs, giving them a sense of achievement. Conversely, this very satisfaction might make it challenging for them to diversify their learning strategies, pushing them toward an inadvertent dependency (Rane et al.; Salah, Al Halbusi, et al., 2023).

This pattern can be juxtaposed with the modern-day phenomenon of 'instant gratification,' where quick answers and validations can sometimes overshadow the deeper, more comprehensive learning processes. Being immediate and precise, Chatbots might be feeding into this very paradigm, thus reinforcing the observed inability to reduce use (Ifelebuegu et al., 2023; Ramirez, 2021).

The broader academic milieu further complicates this relationship. Traditional educational resources, peer interactions, and personal introspection are vital facets of the learning process. The question arises: Are students balancing their chatbot interactions with these resources, or is the scale tilting heavily toward automation?

Moreover, it is essential to consider whether educational institutions inadvertently endorse this dependency by heavily integrating chatbots into their curriculums. The onus might also lie partly with educators to ensure that while chatbots are a supplementary tool, they do not overshadow other critical components of holistic education.

In summary, while the correlation between the need for competence and the inability to reduce chatbot use is telling, the nuances beneath this relationship warrant deeper exploration, shedding light on the intricate interplay of technology, intrinsic motivations, and the evolving landscape of education.

7. Need for Relatedness and Inability to Reduce Chatbot Use:

Contrary to expectations, our data did find a significant relationship between the intrinsic need for relatedness and the inability to reduce chatbot usage. This implies that even if students feel a sense of connection or obtain emotional fulfillment from chatbot interactions, this does not directly translate into an overwhelming compulsion to use or over-rely on them continuously.

Several factors might be influencing this outcome:

1. **Diverse Social Interactions**: Even if chatbots offer a semblance of relatedness, students have many platforms and avenues to fulfill their social and emotional needs. They might be turning to chatbots for specific purposes but are also seeking and finding meaningful human connections elsewhere, diluting the compulsive pull of chatbot interactions (Xie & Pentina, 2022; Xie et al., 2023).
2. **Transient Fulfillment**: The satisfaction derived from chatbot interactions might be ephemeral. While a chatbot can offer immediate responses and simulate human-like conversations, the depth and authenticity of human interactions remain unmatched. Over time, students might recognize this difference and inherently limit their dependence on chatbots for emotional or social fulfillment (Adam et al., 2021; De Gennaro et al., 2020).
3. **Awareness of Artificial Nature**: The cognitive realization that chatbots are, at their core, algorithms without genuine emotions or consciousness might play a role. While students can momentarily feel a sense of connection, the underlying knowledge of its artificiality might prevent the formation of deeper emotional bonds that lead to compulsive behavior (Del Prete, 2021; Weber-Guskar, 2021).
4. **Moderation through Education**: Educational institutions might play a role in ensuring balanced use. Through digital literacy programs and awareness campaigns, students could be informed about the pros and cons of over-relying on digital platforms, including chatbots. This informed stance might be acting as a deterrent against compulsive usage (Cen, 2020; Douglas, 2023).
5. **Evolution of Digital Tools**: As digital tools evolve, students might interact with various platforms, of which chatbots are just one element. This diversified interaction pattern ensures no single platform becomes a dominant focus, even if it offers a sense of relatedness (Clarizia et al., 2018; Huang et al., 2022).

The lack of a significant relationship between the need for relatedness and the inability to reduce chatbot interactions underscores the multifaceted nature of human emotional needs and how they navigate the digital realm. While chatbots offer novel interaction opportunities, they are part of a larger ecosystem where human connections, awareness, and diversified digital engagements shape behavior patterns.

8. Autonomy and Inability to Reduce Chatbot Use:

The dynamics between autonomy and digital tool engagement, specifically chatbots, have become more significant in contemporary education. Our study's findings illuminate that the intrinsic need for autonomy strongly predicts the inability to temper chatbot interactions. Delving into the nuances, it seems that students harnessing chatbots as instruments to fulfill their autonomous learning desires encounter substantial challenges when attempting to disconnect or reduce their usage (Srinivasa et al., 2022; Zimmerman, 2018).

One of the critical attractions of chatbots lies in their promise of instantaneity and customization. They cater to the learner's pace, style, and preferences, essentially handing over the reins of the learning journey to the student. This bolsters a learner's sense of control and profoundly embeds the principles of self-direction and independent decision-making, which resonate with the core tenets of autonomy (Kurni et al., 2023).

However, an unintended consequence emerges as students steep themselves in this environment. The more they immerse themselves in the ecosystem where their autonomous needs are constantly met, the more they develop an ingrained habit loop, making any attempt to reduce chatbot interactions feel almost antithetical to their learned behavior. It is akin to tasting the freedom and empowerment of self-directed learning and then finding oneself tethered when disengaging (Daugherty & Wilson, 2018; Tlili et al., 2023).

Furthermore, the allure of autonomy might obscure the underlying risks, casting a shadow over the pressing need to establish boundaries. As students perceive chatbots as a conduit to manifest their autonomous learning aspirations, they may

inadvertently overlook signs of over-dependence or the subtle shifts from productive engagement to compulsive usage (Smith et al., 2022; Theophilou et al., 2023).

This revelation underscores the intricacies of navigating autonomy in the era of digital ubiquity. Autonomy, undoubtedly, is a cornerstone for fostering self-reliant and proactive learners. Nevertheless, as our findings suggest, there is a precipice where unchecked autonomy, facilitated by tools like chatbots, could spiral into patterns of dependence. The onus then falls on educational institutions and stakeholders to strike a harmonious balance – fostering autonomy without letting it pave the way to unintended digital dependencies.

Exploring the "AIholic" trend provides a comprehensive insight into students' multifaceted relationship with chatbots in the educational landscape. Our findings highlight the intricate interplay between students' psychological needs and their dependency on these digital tools. The significant associations between triggers' needs for competence, relatedness, autonomy, and actual chatbot dependency and the inability to reduce chatbot use emphasize digital stimuli's profound impact on modern learners.

This research underscores the need for a balanced approach to integrating chatbots into the educational domain. While these tools undoubtedly offer opportunities for enhancing autonomous and self-directed learning, our findings also shed light on the potential pitfalls of over-reliance. The dual-edged nature of autonomy in the digital age primarily serves as a poignant reminder of the complexities involved.

As we move forward in this digital era, educators and stakeholders must remain vigilant, ensuring that the integration of AI tools, like chatbots, is strategic and mindful. It is imperative to provide students with the resources and guidance necessary to harness the benefits of these tools without succumbing to counterproductive dependencies.

This study provides a foundational understanding that can guide future research, policy decisions, and pedagogical strategies in the ever-evolving educational landscape by bridging the gap between technology and psychology.

Triggers and Risk Perception

Our results confirmed the significant relationship between the occurrence of 'Triggers' and 'Risk Perception on chatbot usage patterns. The reinforced link suggests that when students frequently encounter triggers or cues prompting chatbot interactions and simultaneously have a reduced perception of the risks, they gravitate towards increased reliance. This is reminiscent of classical conditioning in behavioral psychology, where repeated exposure to a stimulus (in this case, triggers for chatbot usage) can lead to an increased and almost automatic response. The lack of perceived risk serves as an enabler, removing potential barriers that might otherwise deter or limit usage. This highlights the importance of recognizing and managing triggers, especially in a digital landscape with notifications and prompts (Ogilvie et al., 2022; Silva et al., 2023).

The Moderation Role of Risk Perception in User-Chatbot Dependency Dynamics

The structural path analysis has unveiled the crucial role that risk perception plays in determining user-chatbot dependency. By delving deeper into these findings, we understand how perceived risks interplay with other factors, influencing the extent and nature of chatbot usage.

1. **Triggers and Risk Perception**: Hypothesis H-9, which was supported, suggests an intriguing phenomenon. Despite acknowledging the potential risks of chatbot interactions, users might still feel compulsive to engage due to specific triggers or underlying motivations. These triggers can potentially override risk awareness, whether from convenience, efficiency, or personal preferences. Thus, users might find themselves in a conundrum where they recognize the potential downsides but are still drawn to the immediate gratification or utility chatbots offer (Rapp et al., 2021; Xie et al., 2023).
2. **Competence, Dependency, and Risk**: The validation of H-11 introduces another layer of complexity. It suggests that individuals who derive a sense of competence from chatbot interactions can, paradoxically, be more susceptible to

continued use even when they perceive risks. This might be attributed to a confidence or belief that their skillset or knowledge enables them to navigate potential pitfalls more effectively than others (Rajaobelina et al., 2021; Toader et al., 2019).

3. **Relatedness, Dependency, and Risk**: With H-13 and H-14 being supported, we observe a profound impact of 'relatedness' on chatbot usage patterns. Users who resonate or feel a connection with chatbots might be willing to downplay or overlook certain risks because of the intrinsic value they derive from such interactions. The emotional or psychological bond can sometimes overshadow logical or risk-based assessments, emphasizing emotions' profound role in technological interactions (Christoforakos et al., 2021; Nißen et al., 2022).
4. **Autonomy, Dependency, and Risk**: Autonomy presents a mixed bag. While H-15 suggests that risk perceptions less deter autonomy-seeking users, H-16 contradicts this by indicating no significant relationship between autonomy and actual dependency when risks are considered. This nuanced finding hints at a broader narrative where the quest for autonomy might make users more resilient to perceived risks, but it does not necessarily push them into a state of dependency (Kymlicka, 2017; Xie et al., 2023).

In conclusion, risk perception is a double-edged sword in user-chatbot dynamics. On one hand, it serves as a protective mechanism, cautioning users about potential challenges. Conversely, it can inadvertently solidify certain behaviors as users weigh short-term gains against future repercussions. As chatbots become increasingly integrated into our daily lives, understanding this delicate balance becomes paramount for users and developers to foster beneficial and sustainable interactions.

Delving deeper into Table 5, we observe that among the various hypotheses examined, H-12 and H-16 emerge as exceptions in the pattern, with both not gaining the necessary support. This requires an in-depth exploration to truly capture the essence of the factors driving these outcomes and the broader implications for human-chatbot interaction.

Hypothesis H-12 (Competence-> Risk Perception -> Actual Dependency Level) - Not Supported:
This hypothesis suggested a potential interplay between an individual's competence and risk perception influencing their dependency on chatbots. The fact that this interrelation did not hold merit brings forth several intriguing perspectives.

One must ponder whether the conventional understanding of competence—often perceived as an inherent drive to excel or master tasks—might manifest differently when juxtaposed with technological platforms like chatbots. Does perceived competence in one's ability to use a chatbot not intensify one's usage even when one discerns potential risks?

It is entirely possible that users who perceive themselves as competent harness a broad toolkit of strategies and skills. This diversified arsenal could temper the potential impact of risk perception on actual dependency levels (Chandra et al., 2022; Song et al., 2022).

Furthermore, while competence might inherently drive users to maximize their interaction effectiveness, it might not necessarily correlate with prolonged or intensified usage. Competent users might achieve their objectives in a shorter span, reducing their overall interaction time, regardless of their risk perception (Kosch et al., 2023; Pitardi & Marriott, 2021).

Hypothesis H-16 (Autonomy -> Risk Perception -> Actual Dependency Level) - Not Supported:
This hypothesis carved out a space where autonomy and risk perception intersected, anticipating a discernible impact on actual dependency levels. The absence of support for this hypothesis propels us to explore autonomy in the digital interaction realm.

It is pivotal to question whether individuals who cherish autonomy, typically inclined to be independent thinkers and decision-makers, might already possess a discerning perspective toward technology. When introduced to perceived risks, such users might not significantly alter their interaction patterns, reflecting a balanced and intentional approach (Araujo et al., 2020; Neri & Cozman, 2020).

Moreover, the very nature of autonomous individuals might make them less malleable to the external forces of perceived risks. Their engagement patterns with chatbots could be deeply entrenched in personal motivations and perceived benefits rather than swayed by potential hazards (H. Jiang et al., 2022; Mostafa & Kasamani, 2022).

Prior experiences, user goals, and socio-cultural contexts could also impact the interplay between autonomy and risk perception. An autonomous individual with past negative experiences might prioritize perceived risks differently than someone without such experiences (Sartori & Theodorou, 2022; Yadav et al., 2019).

The non-conformity of hypotheses H-12 and H-16 with the anticipated outcomes is a testament to human-chatbot dynamics' intricate and multifaceted nature. It is a stark reminder that while overarching trends and patterns can be identified, individual nuances and subtleties play a monumental role. These insights challenge prevailing notions and pave the way for more nuanced research endeavors. By recognizing and understanding these complexities, we can cultivate a richer and more comprehensive understanding of the evolving human-machine symbiosis, guiding future innovations and interventions in this domain.

Future Research

It would be valuable for future research to cast a wider net regarding student demographics. This could encompass students from a broader age range, different educational stages, or even students with diverse cultural or regional backgrounds. Such an approach would potentially offer more prosperous, more generalized insights. A longitudinal study design could also be considered better to understand the evolution of student behaviors and perceptions.

Another intriguing avenue for future exploration is the diverse functionalities of chatbots. With a wide array of features available in chatbots, it would be insightful to understand how specific features influence student behaviors and meet their psychological needs. While our study was firmly rooted in the Self-Determination Theory, integrating other psychological theories in subsequent studies might provide a more layered understanding of the dynamics at play. For instance, theories such as the Theory of Planned Behavior or Cognitive Load Theory could be woven into the research fabric.

Lastly, future researchers might contemplate adopting qualitative methodologies to capture the nuances of student experiences with chatbots. Techniques such as interviews or focus group discussions could unveil deeper, more personal insights into how students perceive and interact with chatbots in their learning journeys.

Declarations

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Tables

Table 1. Measurement Model, Loading, Construct Reliability and Convergent Validity

Constructs	Items	Loading (> 0.5)	CA (> 0.7)	CR (> 0.7)	AVE (> 0.5)
Triggers	TRG1	0.787	0.814	0.860	0.666
	TRG2	0.856			
	TRG3	0.880			
	TRG4	0.878			
	TRG5	0.815			
	TRG6	0.788			
	TRG7	0.789			
	TRG8	0.798			
	TRG9	0.776			
	TRG10	0.856			
	TRG11	0.640			
	TRG12	0.788			
	TRG13	0.768			
	TRG14	0.856			
	TRG15	0.790			
	TRG16	0.778			
	TRG17	0.758			
	TRG18	0.856			
	TRG19	0.781			
	TRG20	0.746			
Competence	COM1	0.793	0.822	0.882	0.665
	COM2	0.853			
	COM3	0.723			
	COM4	0.772			
	COM5	0.829			
	COM6	0.790			
	COM7	0.753			
Relatedness	REL1	0.733	0.771	0.874	0.687
	REL2	0.769			
	REL3	0.831			
	REL4	0.819			
	REL5	0.781			

	REL6	0.798			
	REL7	0.852			
Autonomy	AUT1	0.784	0.784	0.884	0.659
	AUT2	0.764			
	AUT3	0.810			
	AUT4	0.761			
	AUT5	0.789			
	AUT6	0.852			
	AUT7	0.863			
Risk Perception	RP1	0.641	0.808	0.874	0.651
	RP2	0.780			
	RP3	0.631			
	RP4	0.740			
	RP5	0.803			
	RP6	0.736			
	RP7	0.741			
	RP8	0.759			
The inability to Reduce the Use	IBRU1	0.670	0.814	0.912	0.614
	IBRU2	0.748			
	IBRU3	0.814			
	IBRU4	0.743			
	IBRU5	0.797			
	IBRU6	0.726			
	IBRU7	0.728			
	IBRU8	0.858			
	IBRU9	0.795			
	IBRU10	0.734			
	IBRU11	0.848			
	IBRU12	0.858			
	IBRU13	0.841			
	IBRU14	0.796			

Actual Dependency Level	ADL1	0.801	0.785	0.863	0.586
	ADL2	0.812			
	ADL3	0.795			
	ADL4	0.834			
	ADL5	0.748			
	ADL6	0.758			
	ADL7	0.843			
	ADL8	0.767			
	ADL9	0.763			
	ADL10	0.787			

Notes: CA=Cronbach's Alpha, CR= Composite Reliability, AVE= Average Variance Extracted, the low value e.g., TRG11 and IBRU1 can be retained (Hiar et al., 2017).

Table 2. Descriptive Statistics, Correlation Matrix, and Discriminant Validity Via Fornell and Larcher.

Constructs	Mean	SD	1	2	3	4	5	6	7	8	9	10
1. Triggers	3.855	0.585	**0.778**									
2. Competence	3.792	0.551	0.602	**0.713**								
3. Relatedness	4.168	0.637	0.432	0.514	**0.829**							
4. Autonomy	4.502	0.519	0.134	0.165	0.483	**0.809**						
5. Risk Perception	4.175	0.549	0.335	0.583	0.472	0.305	**0.742**					
6. The inability to Reduce the Use	4.133	0.521	0.408	0.574	0.555	0.248	0.602	**0.717**				
7. Actual Dependency Level	1.238	0.426	0.326	0.130	0.141	0.246	0.121	0.122	**0.621**			
8. Age	3.044	1.004	-0.092	-0.047	-0.007	0.058	0.035	-0.095	-0.024	n.a		
9. Education	2.828	1.216	0.011	0.003	-0.059	-0.114	-0.028	-0.078	0.025	0.222	n.a	
10. Job Experience	3.281	1.197	-0.167	-0.079	0.074	0.083	0.038	-0.019	-0.021	0.694	0.145	n.a

Notes: S.D. = Standard Deviation. n.a= not applicable. Bold values on the diagonal in the correlation matrix are square roots of AVE (variance shared between the constructs and their respective measures). Off-diagonal elements below the diagonal are correlations among the constructs, where values between 0.13 and 0.16 are significant at $p<0.05$, and values above 0.16 are significant at $p < 0.01$ (two-tailed test).

Table 3. Measurement Model, Discriminant Validity via (HTMT Criterion)

Constructs	1	2	3	4	5	6	7
1. Triggers							
2. Competence	0.560						
3. Relatedness	0.623	0.658					
4. Autonomy	0.549	0.922	0.605				
5. Risk Perception	0.529	0.479	0.874	0.392			
6. The inability to Reduce the Use	0.579	0.391	0.434	0.403	0.392		
7. Actual Dependency Level	0.750	0.615	0.652	0.660	0.536	0.643	
Notes: HTMT should be lower than 0.85.							

Table 4. Structural Path Analysis: Direct Effect.

						Bias and Corrected Bootstrap 95% CI	
Hypothesis	**Relationship**	**Std Beta**	**Std Error**	**t-value**	**p-values**	**[Lower Level; Upper Level]**	**Decision**
H-1	Triggers -> Inability to Reduce Use	0.386	0.060	6.467	0.000	[0.275; 0.475]	Supported
H-2	Triggers -> Actual Dependency Level	0.198	0.066	3.019	0.001	[0.075; 0.293]	Supported
H-3	Competence -> Inability to Reduce Use	0.280	0.070	3.990	0.000	[0.166; 0.404]	Supported
H-4	Competence -> Actual Dependency Level	0.301	0.062	4.881	0.000	[0.187; 0.396]	Supported
H-5	Relatedness-> Inability to Reduce Use	0.154	0.045	3.412	0.000	[0.155; 0.341]	Supported
H-6	Relatedness-> Actual Dependency Level	0.101	0.062	1.024	0.153	[-0.117; 0.020]	Not Supported
H-7	Autonomy-> Inability to Reduce Use	0.128	0.049	2.583	0.005	[0.154; 0.412]	Supported
H-8	Autonomy-> Actual Dependency Level	0.242	0.052	4.672	0.000	[0.153; 0.381]	Supported
Note (s): n=366. Bootstrap sample size = 5,000. SE=standard error; LL=lower limit; CI=confidence interval; UL=upper limit 95% bias-correlated CI.							

Table 5. Structural Path Analysis: The Interaction Effect.

						Bias and Corrected Bootstrap 95% CI	
Hypothesis	**Relationship**	**Std Beta**	**Std Error**	**t-value**	**p-values**	**[Lower Level; Upper Level]**	**Decision**
H-9	Triggers * Risk Perception-> Inability to Reduce Use	0.066	0.025	2.641	0.001	[0.023; 0.104]	Supported
H-10	Triggers * Risk Perception -> Actual Dependency Level	0.215	0.061	3.524	0.000	[0.124; 0.321]	Supported
H-11	Competence * Risk Perception -> Inability to Reduce Use	0.049	0.018	2.722	0.003	[0.020;0.076]	Supported
H-12	Competence * Risk Perception -> Actual Dependency Level	0.106	0.093	1.139	0.126	[0.010; -0.274]	Not Supported
H-13	Relatedness * Risk Perception -> Inability to Reduce Use	0.099	0.046	2.166	0.001	[0.027;0.176]	supported
H-14	Relatedness * Risk Perception -> Actual Dependency Level	0.131	0.052	4.640	0.000	[0.041; 0.203]	Supported
H-15	Autonomy * Risk Perception -> Inability to Reduce Use	0.149	0.050	2.988	0.001	[0.064; 0.232]	Supported
H-16	Autonomy * Risk Perception -> Actual Dependency Level	0.045	0.044	1.024	0.153	[-0.117; 0.020]	Not supported

Note (s): n=366. Bootstrap sample size = 5,000. SE=standard error; LL=lower limit; CI=confidence interval; UL=upper limit 95% bias-correlated CI.

Figures

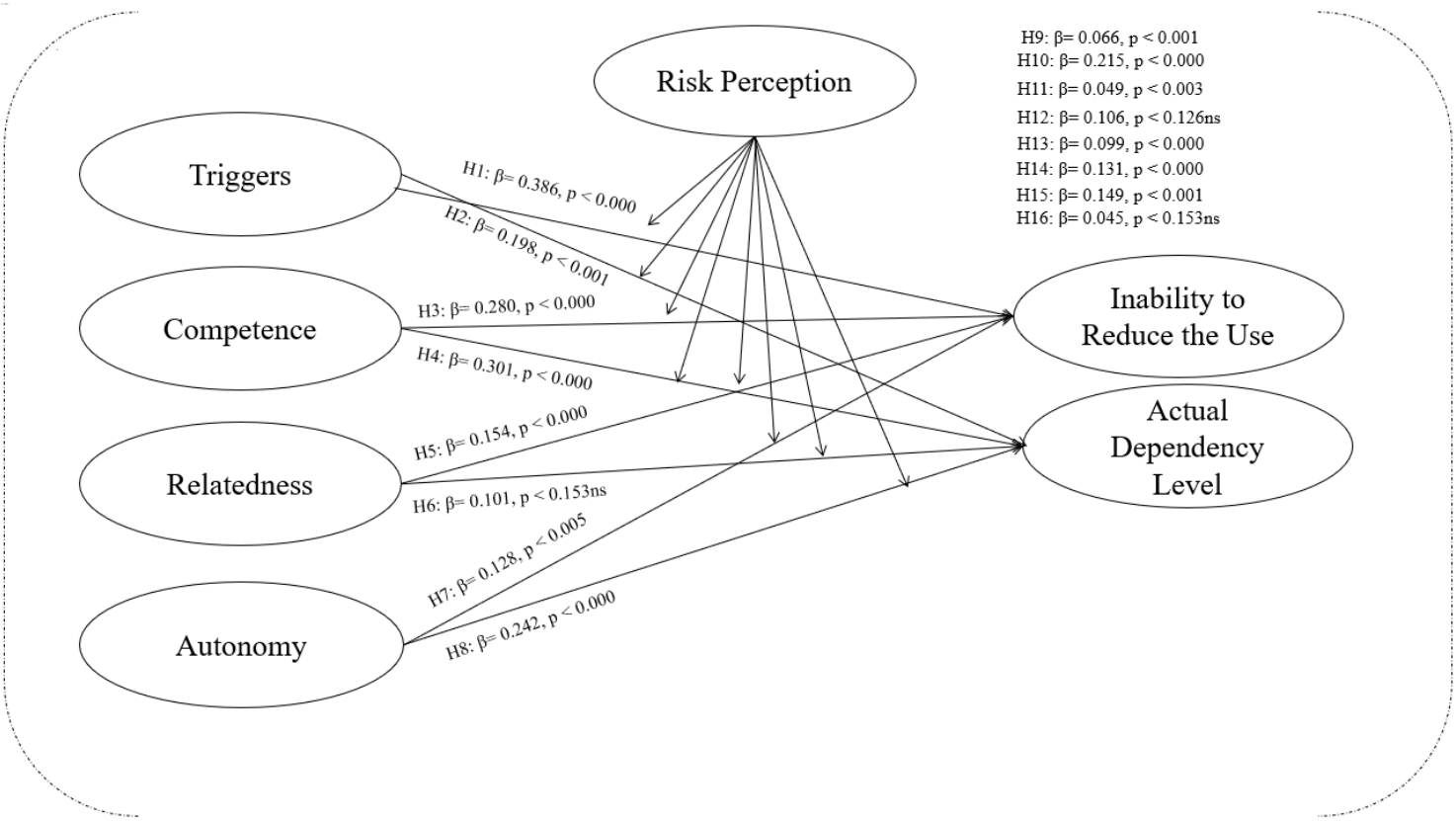

Figure 1

Figure 2. Hypotheses Testing

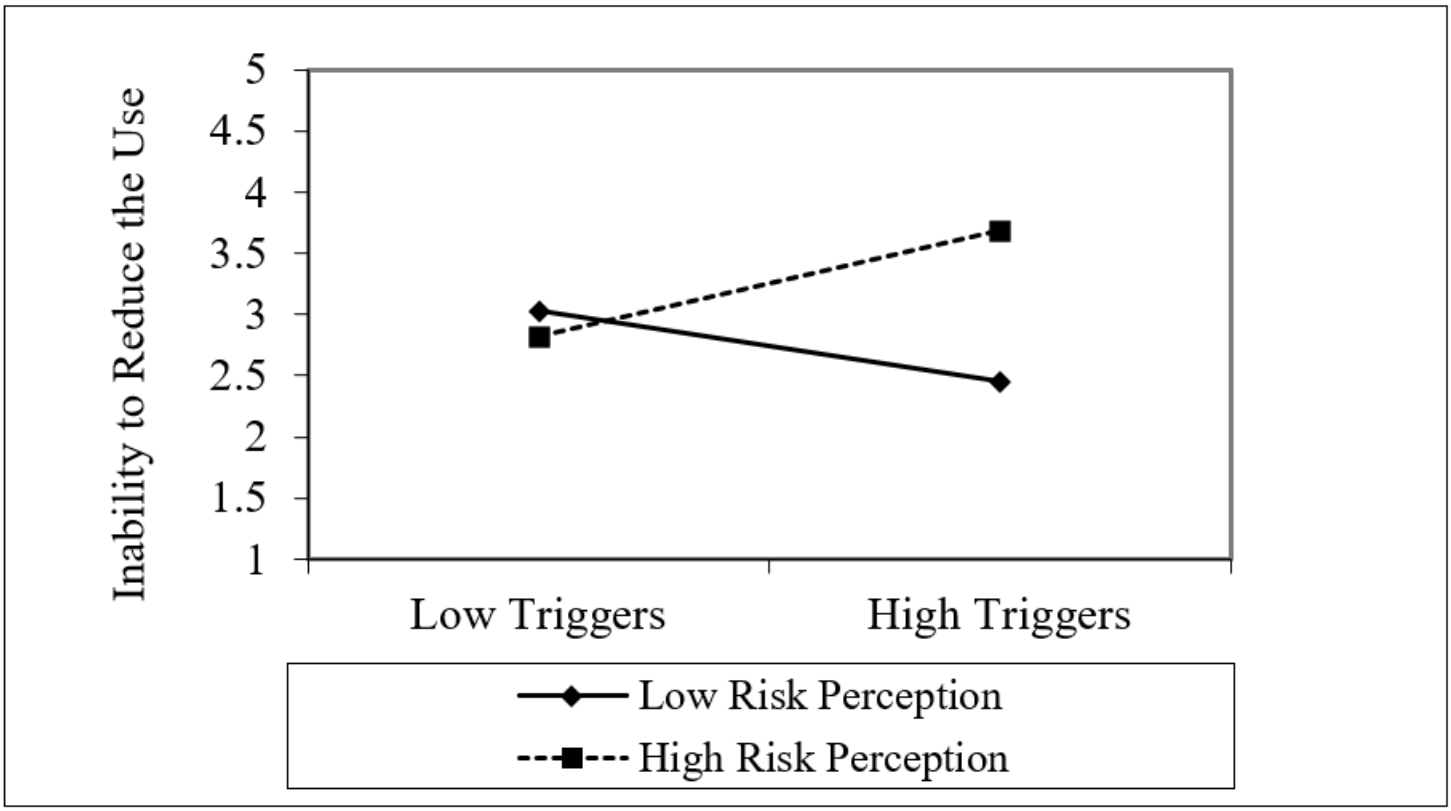

Figure 2

Figure 3. Interaction of triggers and risk perception on the inability to reduce the use

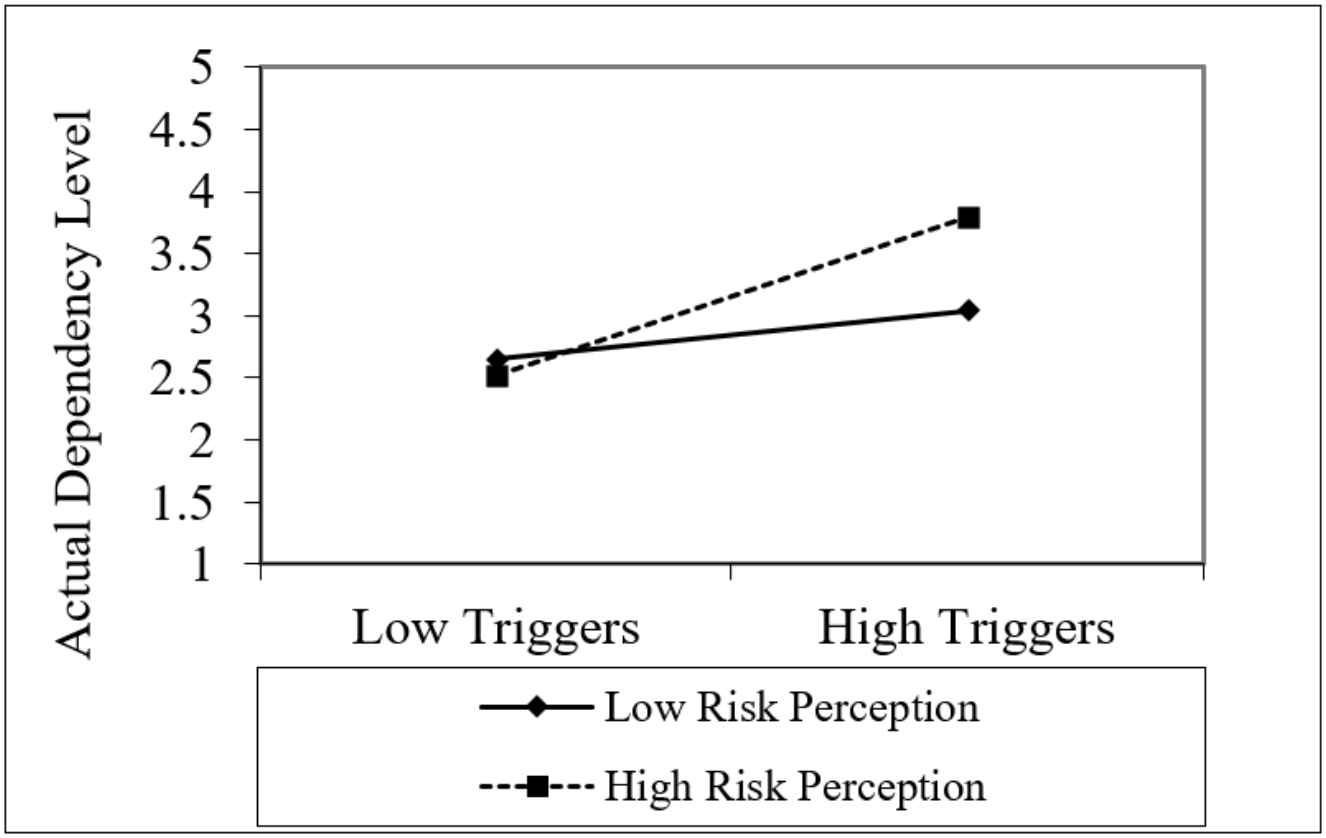

Figure 3

Figure 4. Interaction of triggers and risk perception on the actual dependency level

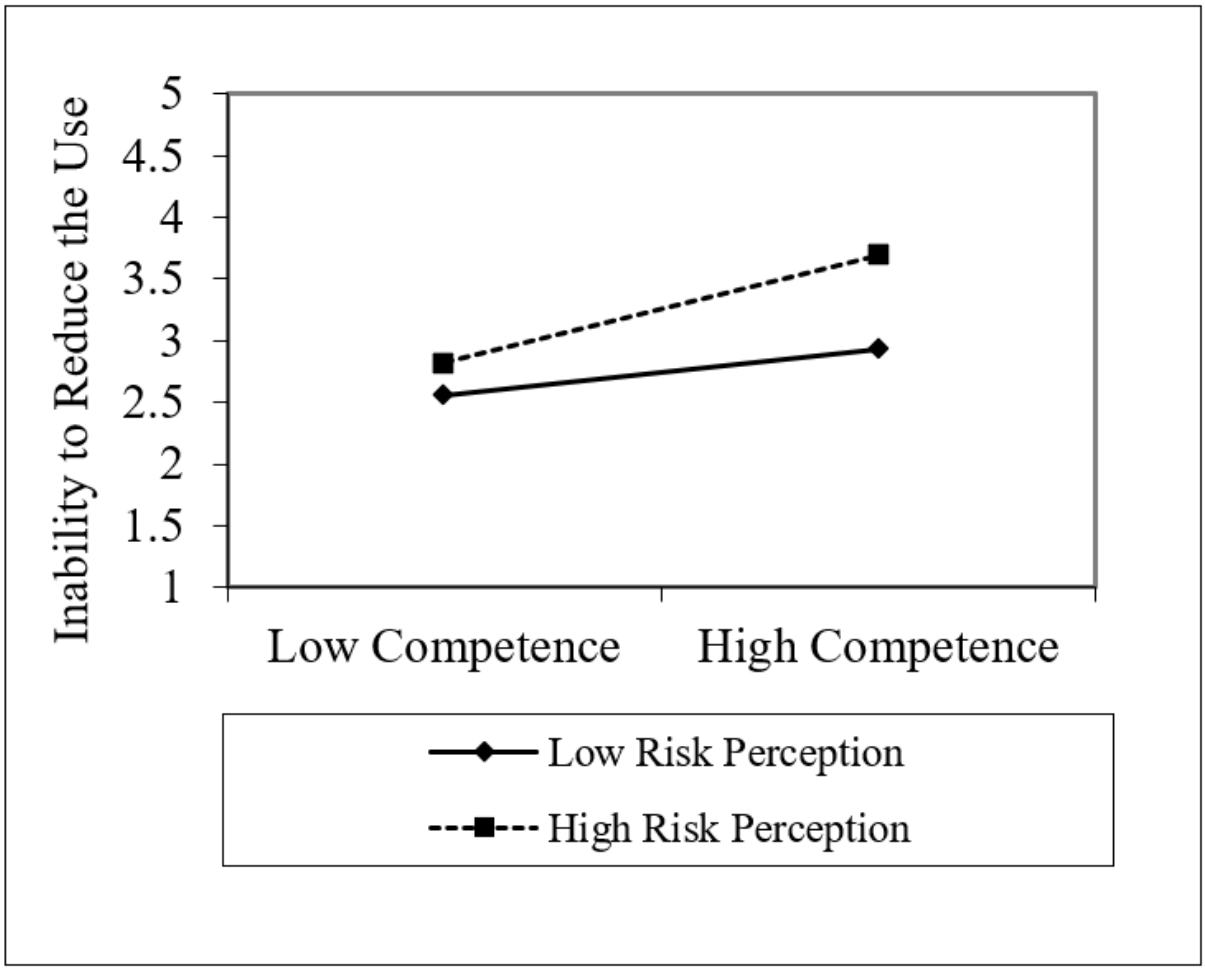

Figure 4

Figure 5. Interaction of Competence and risk perception on the Inability to Reduce the Use

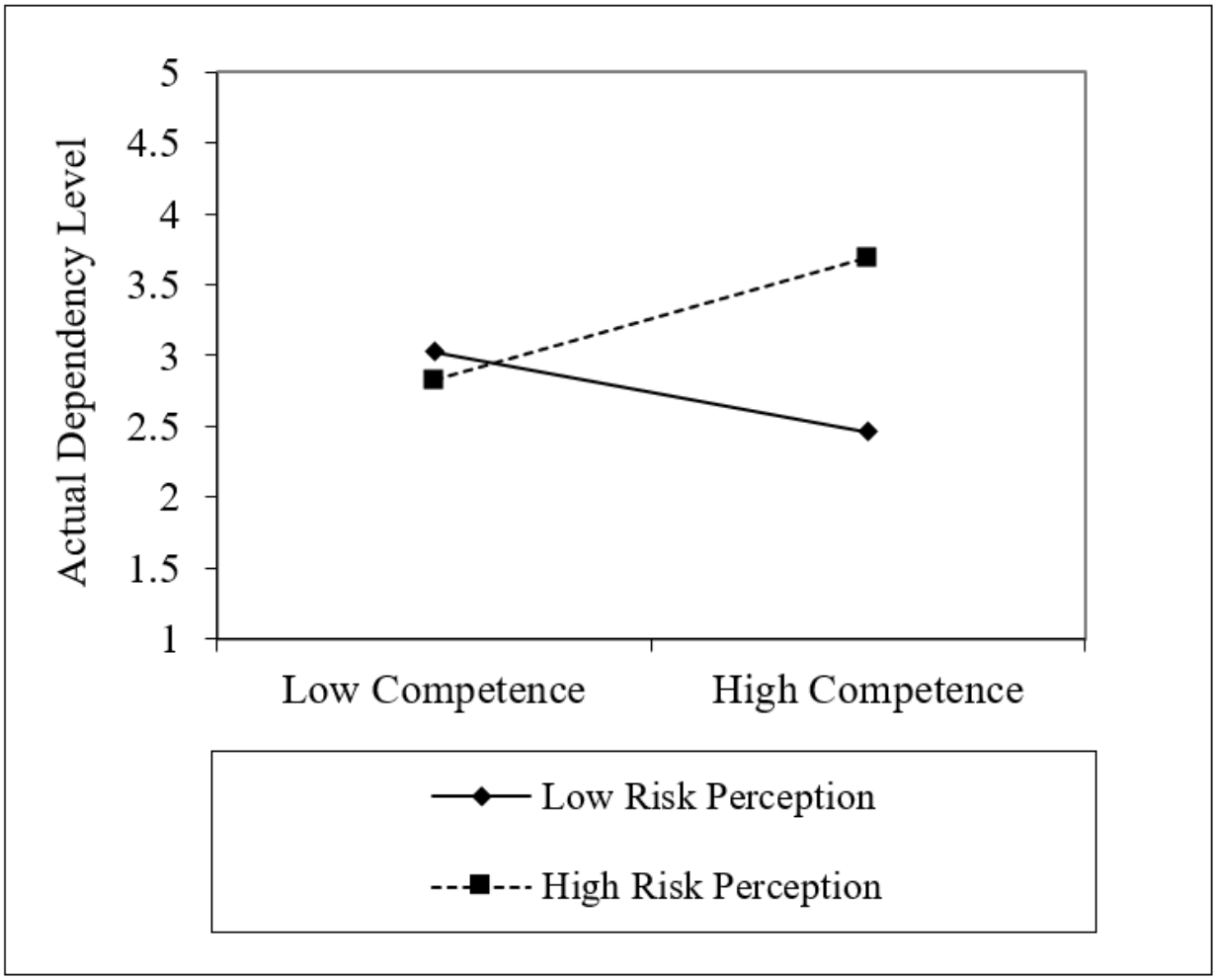

Figure 5

Figure 6. Interaction of Competence and risk perception on the actual dependency level

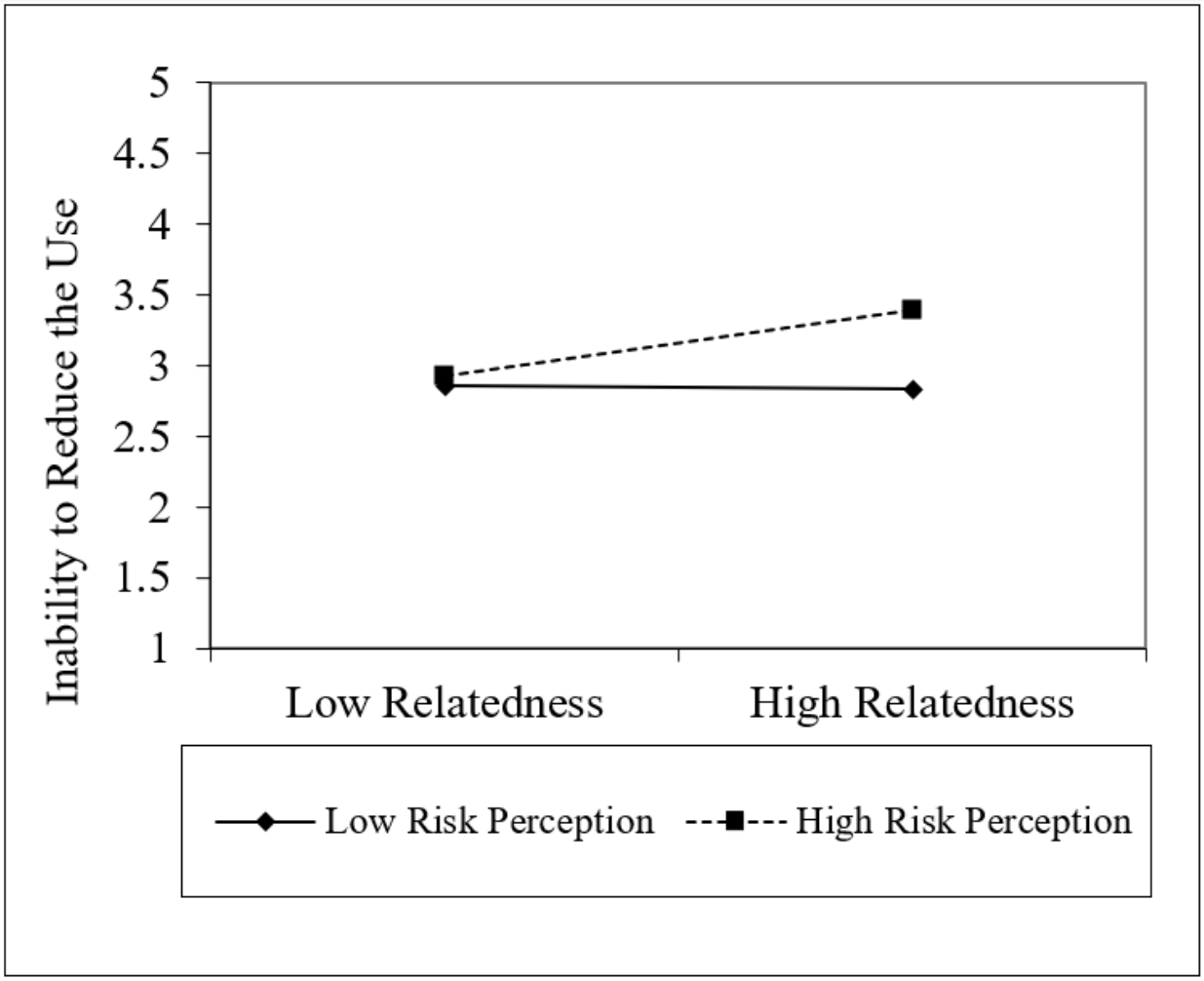

Figure 6

Figure 7. Interaction of relatedness and risk perception on the inability to reduce the use

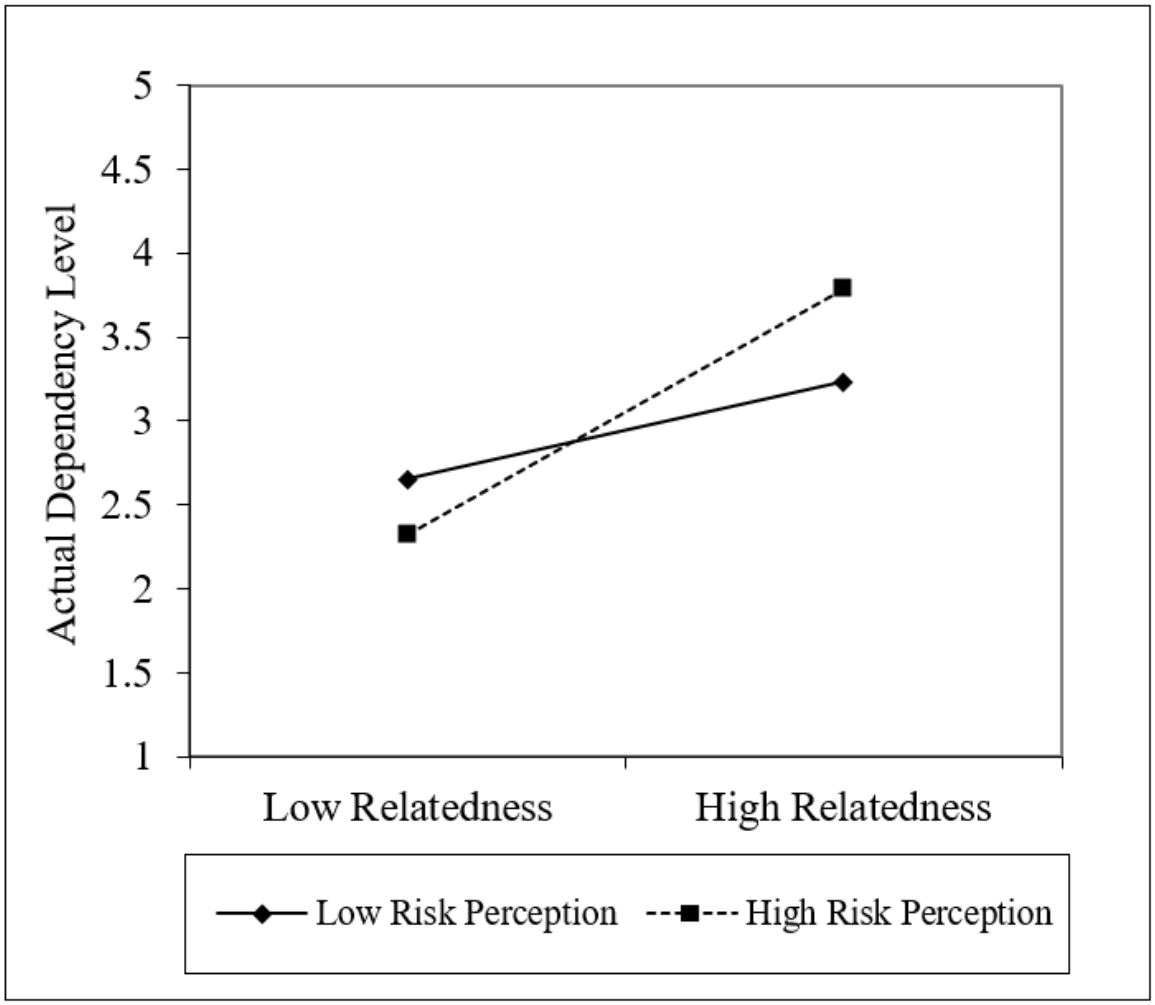

Figure 7

Figure 8. Interaction of relatedness and risk perception on the actual dependency level

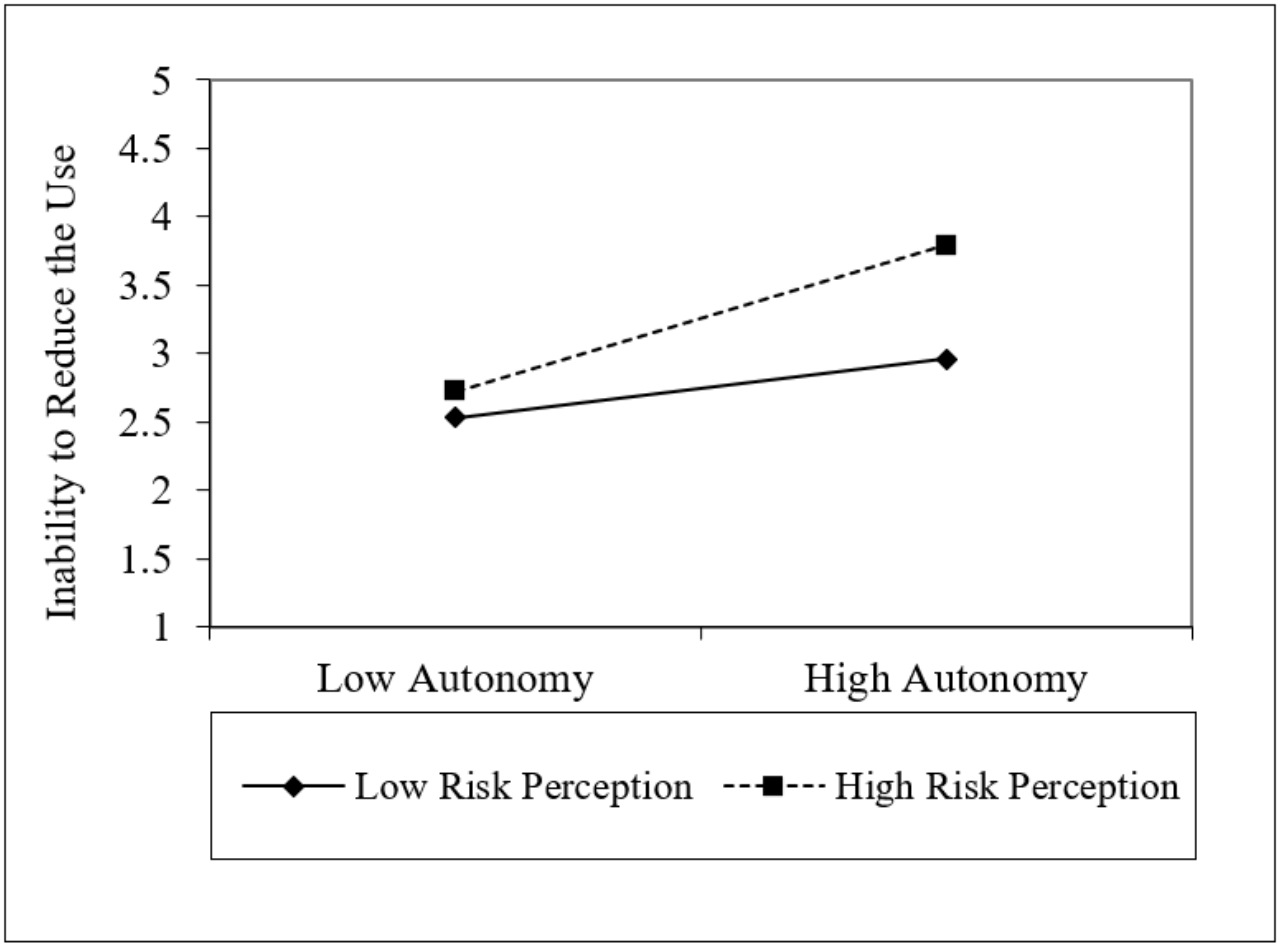

Figure 8

Figure 9. Interaction of Autonomy and risk perception on the inability to reduce the use

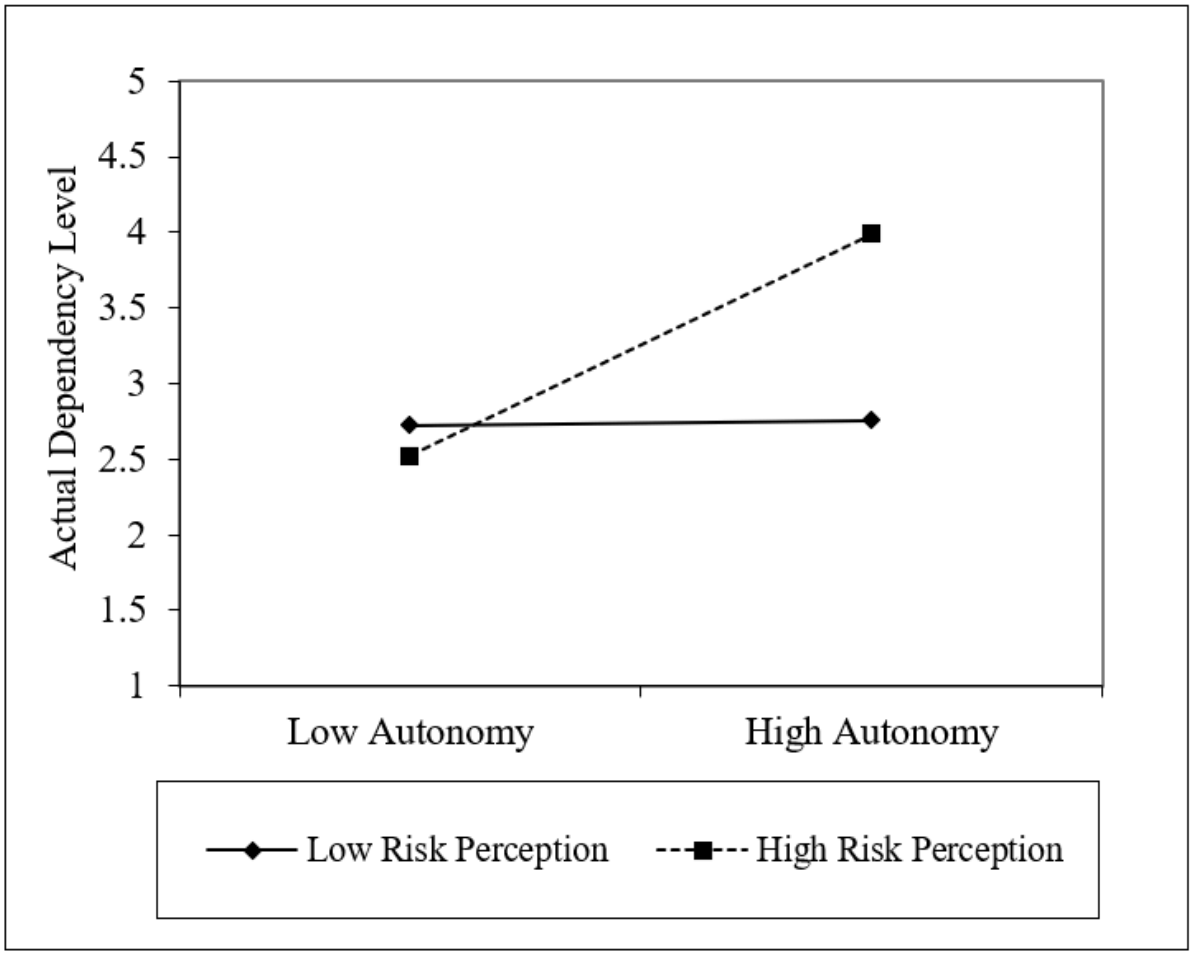

Figure 9

Figure 10. Interaction of Autonomy and risk perception on the actual dependency level

Supplementary Files

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